SC 06 · WHEN THINGS BREAK

Detect, isolate, recover

A chipped prop, a dead ESC: what a quadrotor can and cannot survive, using the rotorfault simulation study.

Loss of effectiveness

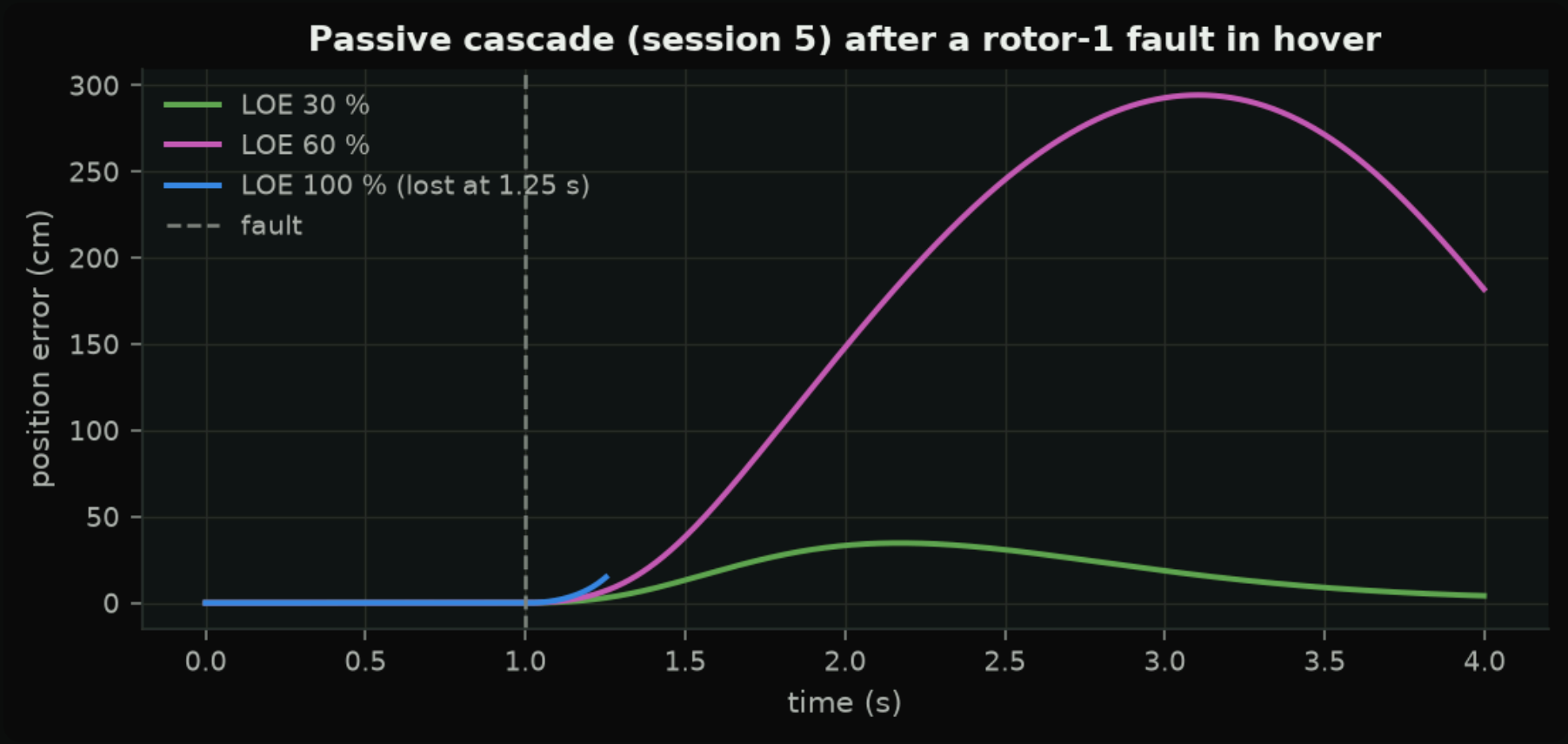
$$\begin{bmatrix} T \\ \tau \end{bmatrix} = B\,\mathrm{diag}(\eta)\,f, \qquad \eta_j \in [0,1]$$

Hover with heading held needs every rotor to give $mg/4$:

$$\eta_j \geq \frac{1}{T/W} \qquad (= 0.28 \text{ for the study's quad})$$

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A passive controller



Integral action absorbs a moderate loss; a complete loss flips it in ~0.3 s.

Residuals and signatures

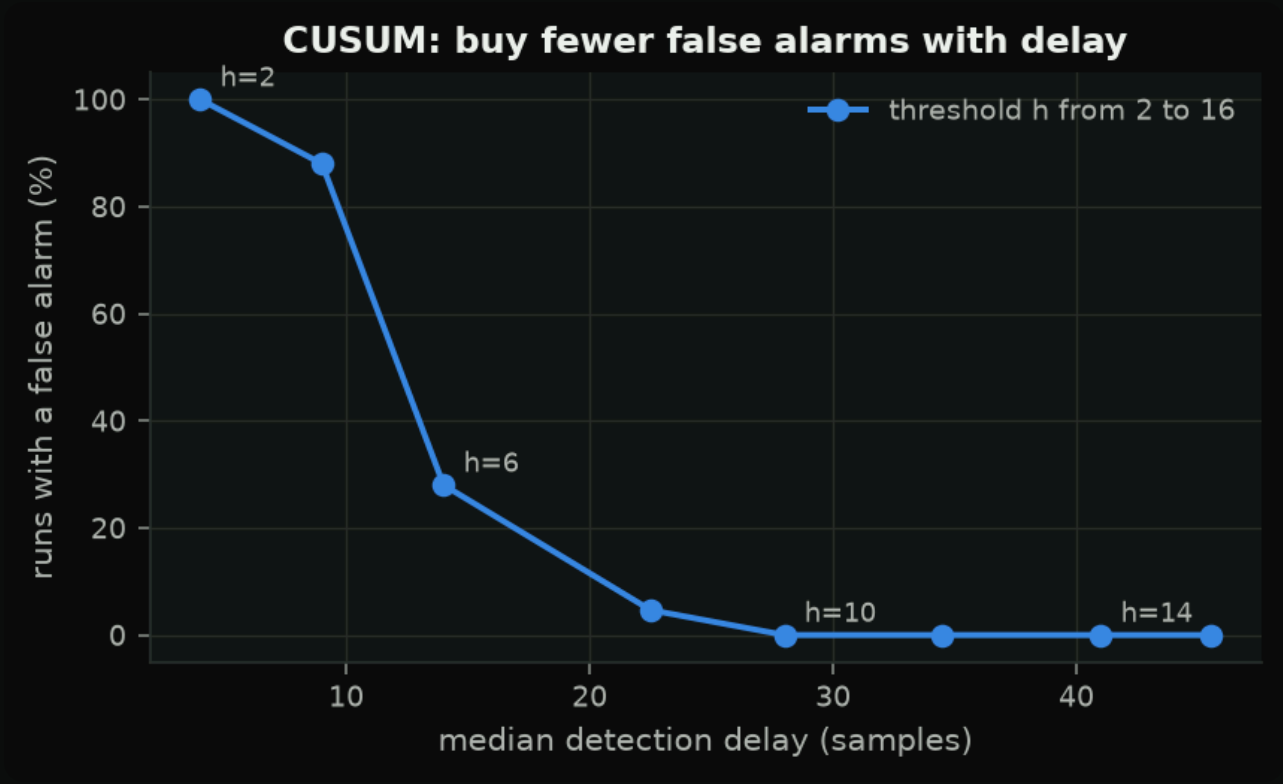
$$r = -\delta_j B_{:,j} f_j + \text{noise} = \delta_j d_j + \text{noise}, \qquad \hat{j} = \arg\max_j \frac{d_j^T r}{\|d_j\|}$$

Each rotor pushes the residual along a known direction.

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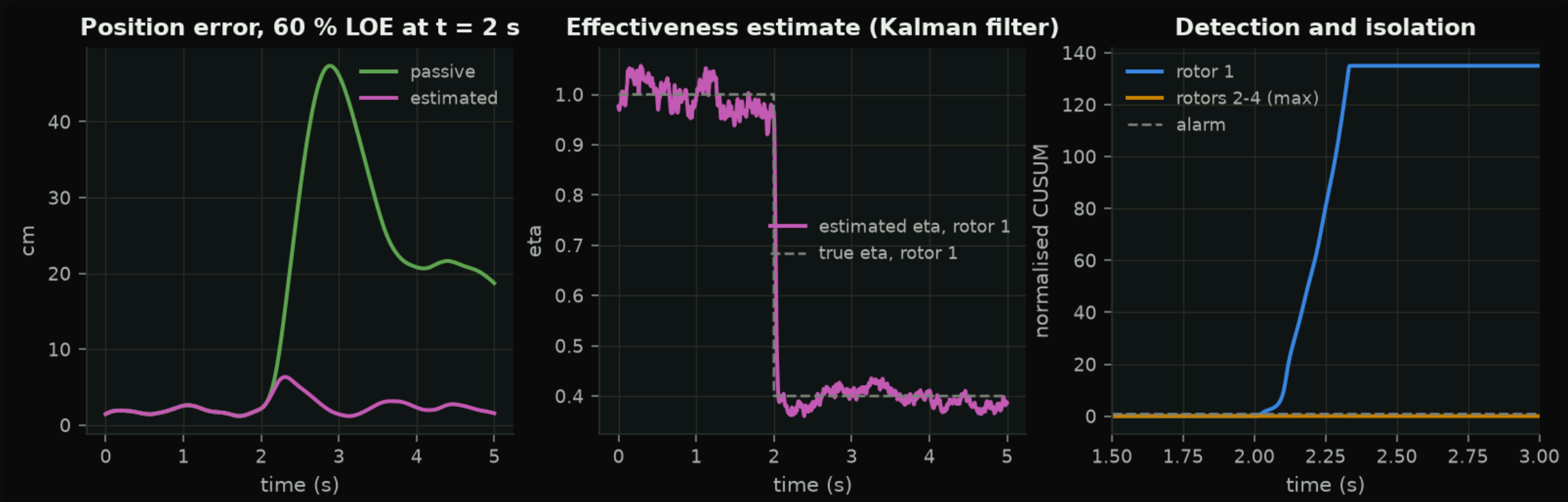
CUSUM: evidence over time

$$g_k = \max\left(0,\ g_{k-1} + \frac{\mu_1}{\sigma^2}\left(r_k - \frac{\mu_1}{2}\right)\right) > h$$



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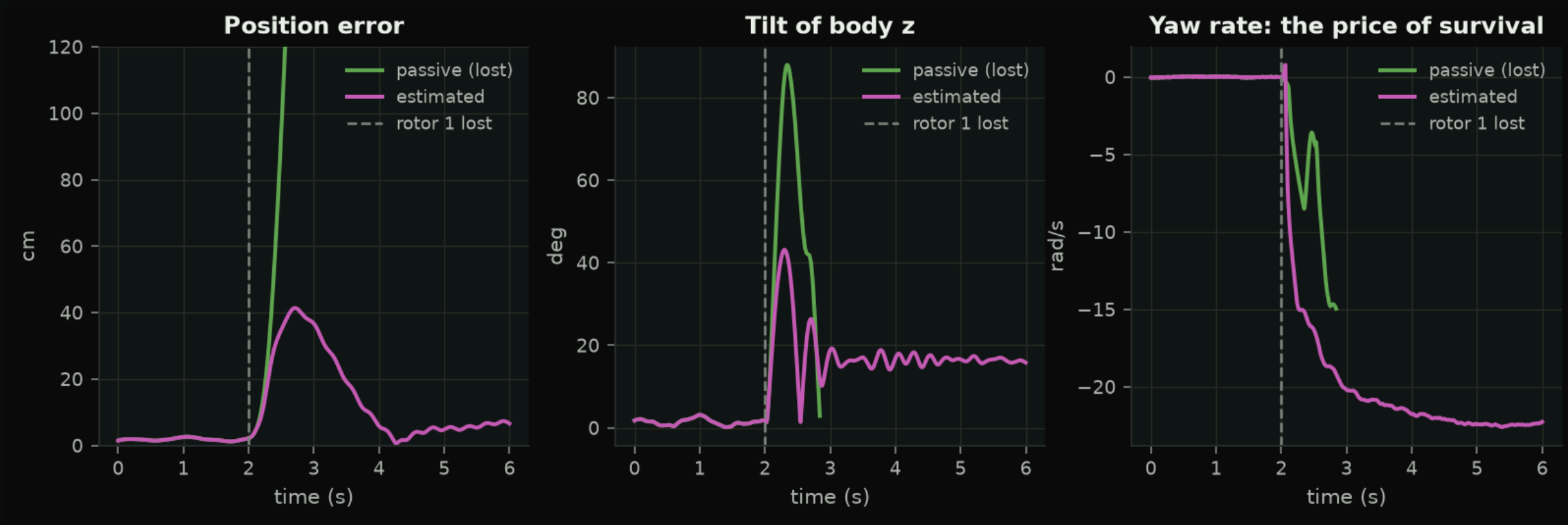
The rotorfault pipeline, 60 % loss



Estimator → filtered regression → Kalman filter on η → CUSUM bank → supervisor → re-allocation with $B\,\mathrm{diag}(\hat{\eta})$.

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Complete loss: give up yaw



Spin about a tilted primary axis (Mueller & D'Andrea, 2014); control its direction and the thrust. 24 rad/s, inside the gyro range.

Read the limits first

From the study's own section 6:

- simulation only, **assumed** parameters
- white-noise vibration, motion capture, one clean step fault
- recovery **failed** at 24 ms actuator delay, 80 ms motor lag, 3x turbulence

A good study tells you where it breaks.

Lab 6

- **A** rotorfault: loss sweep, detection delay, false alarms, the motor-lag limit
- **B** (optional, rig) `fault 0.7` and your own CUSUM on the controller effort

Then: capstone teams and tracks.